Now calculate the values of $\frac{dS_1}{d\,l}$ for the case when $l \to 0$ and k is sufficiently large.

$$
\begin{aligned}
&\lim_{l\to 0}\frac{dS_1}{d\,l}\\
=&\alpha^3c^{3\alpha}\lim_{l\to 0}\int_c^{+\infty}\int_c^{+\infty}\int_c^{+\infty}\frac{(x_1+x_2+x_3)}{3x_1^{\alpha+1}x_2^{\alpha+1}x_3^{\alpha+1}}\\
&\cdot\frac{(k+1+l)x_3-(\frac{k}{3}(x_1+x_2+x_3)+x_2+lx_3)}{(k+1+l)(\frac{k}{3}(x_1+x_2+x_3)+x_2+lx_3)}dx_1dx_2dx_3\\
=&\alpha^3c^{3\alpha}\lim_{l\to 0}\int_c^{+\infty}\int_c^{+\infty}\int_c^{+\infty}\frac{(x_1+x_2+x_3)x_3}{3x_1^{\alpha+1}x_2^{\alpha+1}x_3^{\alpha+1}(\frac{k}{3}(x_1+x_2+x_3)+x_2+lx_3)}dx_1dx_2dx_3\\
&-\alpha^3c^{3\alpha}\lim_{l\to 0}\int_c^{+\infty}\int_c^{+\infty}\int_c^{+\infty}\frac{(x_1+x_2+x_3)}{3x_1^{\alpha+1}x_2^{\alpha+1}x_3^{\alpha+1}(k+1+l)}dx_1dx_2dx_3\\
=&\alpha^3c^{3\alpha}\int_c^{+\infty}\int_c^{+\infty}\int_c^{+\infty}\lim_{l\to 0}\frac{(x_1+x_2+x_3)x_3}{3x_1^{\alpha+1}x_2^{\alpha+1}x_3^{\alpha+1}(\frac{k}{3}(x_1+x_2+x_3)+x_2+lx_3)}dx_1dx_2dx_3\\
&-\alpha^3c^{3\alpha}\lim_{l\to 0}\int_c^{+\infty}\int_c^{+\infty}\int_c^{+\infty}\frac{x_1}{x_1^{\alpha+1}x_2^{\alpha+1}x_3^{\alpha+1}(k+1+l)}dx_1dx_2dx_3\\
=&\alpha^3c^{3\alpha}\int_c^{+\infty}\int_c^{+\infty}\int_c^{+\infty}\frac{x_1+x_2+x_3}{3x_1^{\alpha+1}x_2^{\alpha+1}x_3^{\alpha+1}}\\
&\cdot\lim_{l\to 0}\left[\frac{x_3}{\frac{k}{3}(x_1+x_2+x_3)+x_2}-\frac{x_3^2}{[\frac{k}{3}(x_1+x_2+x_3)+x_2]^2}l\right]dx_1dx_2dx_3\\
&-\alpha^3c^{3\alpha}\lim_{l\to 0}\int_c^{+\infty}\int_c^{+\infty}\int_c^{+\infty}\frac{x_1}{x_1^{\alpha+1}x_2^{\alpha+1}x_3^{\alpha+1}(k+1+l)}dx_1dx_2dx_3 \qquad (8)\\
=&\alpha^3c^{3\alpha}\int_c^{+\infty}\int_c^{+\infty}\int_c^{+\infty}\frac{x_1+x_2+x_3}{3x_1^{\alpha+1}x_2^{\alpha+1}x_3^{\alpha+1}}\frac{x_3}{\frac{k}{3}(x_1+x_2+x_3)+x_2}dx_1dx_2dx_3\\
&-\alpha^3c^{3\alpha}\lim_{l\to 0}\int_c^{+\infty}\int_c^{+\infty}\int_c^{+\infty}\frac{x_1}{x_1^{\alpha+1}x_2^{\alpha+1}x_3^{\alpha+1}(k+1+l)}dx_1dx_2dx_3\\
=&\alpha^3c^{3\alpha}\int_c^{+\infty}\int_c^{+\infty}\int_c^{+\infty}\frac{x_1+x_2+x_3}{3x_1^{\alpha+1}x_2^{\alpha+1}x_3^{\alpha+1}}\left[\frac{x_3}{(x_1+x_2+x_3)}\frac{3}{k}-\frac{x_2x_3}{(x_1+x_2+x_3)^2}\frac{9}{k^2}\right]dx_1dx_2dx_3\\
&-\alpha^3c^{3\alpha}\lim_{l\to 0}\int_c^{+\infty}\int_c^{+\infty}\int_c^{+\infty}\frac{x_1}{x_1^{\alpha+1}x_2^{\alpha+1}x_3^{\alpha+1}(k+1+l)}dx_1dx_2dx_3 \qquad (9)\\
=&\alpha^3c^{3\alpha}\int_c^{+\infty}\int_c^{+\infty}\int_c^{+\infty}\frac{x_1}{x_1^{\alpha+1}x_2^{\alpha+1}x_3^{\alpha+1}}\frac{1}{k(k+1)}dx_1dx_2dx_3\\
&-\alpha^3c^{3\alpha}\int_c^{+\infty}\int_c^{+\infty}\int_c^{+\infty}\frac{1}{x_1^{\alpha+1}x_2^{\alpha+1}x_3^{\alpha+1}}\frac{3x_2x_3}{k^2(x_1+x_2+x_3)}dx_1dx_2dx_3 \qquad (10)
\end{aligned}
$$

Here, we apply a Taylor expansion to Equation (8) and Equation (9) with respect to l and $\frac{1}{k}$, respectively.

Similarly, we can calculate the value of $\frac{dS_2}{d\,l}$.

$$
\begin{aligned}
&\lim_{l\to 0}\frac{dS_2}{d\,l}\\
=&\alpha^3c^{3\alpha}\lim_{l\to 0}\int_c^{+\infty}\int_c^{+\infty}\int_c^{+\infty}\frac{x_2}{x_1^{\alpha+1}x_2^{\alpha+1}x_3^{\alpha+1}}\\
&\cdot\frac{(k+1+l)x_3-(\frac{k}{3}(x_1+x_2+x_3)+x_2+lx_3)}{(k+1+l)(\frac{k}{3}(x_1+x_2+x_3)+x_2+lx_3)}dx_1dx_2dx_3\\
=&\lim_{l\to 0}\int_c^{+\infty}\int_c^{+\infty}\int_c^{+\infty}\frac{\alpha^3c^{3\alpha}}{x_1^{\alpha+1}x_2^{\alpha+1}x_3^{\alpha+1}}\cdot\left(\frac{3x_2x_3}{k(x_1+x_2+x_3)}-\frac{1}{k+1+l}\right)dx_1dx_2dx_3 \qquad (11)
\end{aligned}
$$

As a result, Equation (7) can be written as:

$$
\begin{aligned}
&\lim_{k\to+\infty,\, l\to 0} \frac{k\cdot \frac{d\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_p/\mathcal{D}}}{d\,l} + \frac{d\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_a/\mathcal{D}}}{d\,l}}{k\cdot \frac{d\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_p/\mathcal{D}}}{d\,l}} \\
=&\frac{\lim_{k\to+\infty,\, l\to 0}\left(\frac{k\cdot dS_1+dS_2}{d\,l}\right)}{\lim_{k\to+\infty,\, l\to 0}\frac{k\cdot dS_1}{d\,l}} \\
&\lim_{k\to+\infty,\, l\to 0}\Bigg(k\cdot\int_c^{+\infty}\int_c^{+\infty}\int_c^{+\infty}\frac{\alpha^3c^{3\alpha}}{\prod_{i=1}^3 x_i^{\alpha+1}}\left[\frac{x_1}{k(k+1)}-\frac{3x_2x_3}{k^2(\sum_{i=1}^3 x_i)}\right]dx_1dx_2dx_3 \\
&\qquad\qquad +\int_c^{+\infty}\int_c^{+\infty}\int_c^{+\infty}\frac{\alpha^3c^{3\alpha}}{\prod_{i=1}^3 x_i^{\alpha+1}}\cdot\left(\frac{3x_2x_3}{k(\sum_{i=1}^3 x_i)}-\frac{1}{k+1+l}\right)dx_1dx_2dx_3\Bigg) \\
=&\frac{\lim_{k\to+\infty,\, l\to 0} k\cdot\int_c^{+\infty}\int_c^{+\infty}\int_c^{+\infty}\frac{1}{\prod_{i=1}^3 x_i^{\alpha+1}}\left[\frac{x_1}{k(k+1)}-\frac{3x_2x_3}{k^2(\sum_{i=1}^3 x_i)}\right]dx_1dx_2dx_3}{\lim_{k\to+\infty,\, l\to 0} k\cdot\int_c^{+\infty}\int_c^{+\infty}\int_c^{+\infty}\frac{1}{\prod_{i=1}^3 x_i^{\alpha+1}}\left[\frac{x_1}{k(k+1)}-\frac{3x_2x_3}{k^2(\sum_{i=1}^3 x_i)}\right]dx_1dx_2dx_3} \\
=&\frac{\lim_{k\to+\infty,\, l\to 0}\int_c^{+\infty}\int_c^{+\infty}\int_c^{+\infty}\frac{\alpha^3c^{3\alpha}}{\prod_{i=1}^3 x_i^{\alpha+1}}\cdot\frac{l}{(k+1)(k+1+l)}}{\lim_{k\to+\infty,\, l\to 0} k\cdot\int_c^{+\infty}\int_c^{+\infty}\int_c^{+\infty}\frac{1}{\prod_{i=1}^3 x_i^{\alpha+1}}\left[\frac{x_1}{k(k+1)}-\frac{3x_2x_3}{k^2(\sum_{i=1}^3 x_i)}\right]dx_1dx_2dx_3} \\
=&0.
\end{aligned}
\tag{12}
$$

Next, we prove that $\frac{d\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_1/\mathcal{D}}}{d\,l} < 0$. For sufficiently large k, since

$$
\begin{aligned}
&\lim_{l\to 0}\frac{dS_1}{d\,l} \\
=&\alpha^3c^{3\alpha}\lim_{l\to 0}\int_c^{+\infty}\int_c^{+\infty}\int_c^{+\infty}\frac{x_1}{x_1^{\alpha+1}x_2^{\alpha+1}x_3^{\alpha+1}}\frac{1}{k(k+1)}dx_1dx_2dx_3 \\
&-\alpha^3c^{3\alpha}\lim_{l\to 0}\int_c^{+\infty}\int_c^{+\infty}\int_c^{+\infty}\frac{1}{x_1^{\alpha+1}x_2^{\alpha+1}x_3^{\alpha+1}}\frac{3x_2x_3}{k^2(x_1+x_2+x_3)}dx_1dx_2dx_3 \\
=&\int_c^{+\infty}\int_c^{+\infty}\int_c^{+\infty}\frac{\frac{k}{k+1}x_1(x_1+x_2+x_3)-3x_1x_3}{k(x_1+x_2+x_3)x_1^{\alpha+1}x_2^{\alpha+1}x_3^{\alpha+1}}dx_1dx_2dx_3 \\
:=&\frac{1}{k}\int_c^{+\infty}\int_c^{+\infty}\int_c^{+\infty}\frac{f(x_1,x_2,x_3)}{g(x_1,x_2,x_3)}dx_1dx_2dx_3.
\end{aligned}
\tag{13}
$$

For $c < u < v < w$, let

$$
\begin{aligned}
\hat{f} = x_1(x_1+x_2+x_3)-3x_1x_3 = x_1(x_1+x_2-2x_3),& \\
h(u,v,w) = \sum_{\pi\in S_3} f(\pi(u,v,w)),& \\
\hat{h}(u,v,w) = \sum_{\pi\in S_3} \hat{f}(\pi(u,v,w)).&
\end{aligned}
\tag{14}
$$

Here, S_3 denotes the symmetric group of degree 3 and $\pi \in S_3$ denotes a permutation in S_3. Observe that

$$
\begin{aligned}
\lim_{k\to+\infty} h(u,v,w) =& \hat{h}(u,v,w)^-, \\
\hat{h}(u,v,w) =& \sum_{\pi\in S_3}\hat{f}(\pi(u,v,w)) \\
=& \sum_{\substack{\pi\in S_3 \\ (u_0,v_0,w_0)=\pi(u,v,w)}} u_0(u_0+v_0-2w_0) \\
=& \sum_{\substack{\pi\in S_3 \\ (u_0,v_0,w_0)=\pi(u,v,w)}} \frac{1}{2}(u_0-v_0)^2 > 0.
\end{aligned}
\tag{15}
$$

Hence, we have

$$\lim_{k\to+\infty} h(u,v,w) > 0. \tag{16}$$

Therefore,

$$\lim_{k\to+\infty} \frac{h(u,v,w)}{g(u,v,w)} = \lim_{k\to+\infty} \frac{\Theta(1)}{\Theta(k)} = 0^+. \tag{17}$$

Therefore,

$$\begin{aligned}
\lim_{k\to+\infty, l\to 0} \frac{dS_1}{d\,l} &= \lim_{k\to+\infty} \iiint_{(u,v,w)\in(c,+\infty)^3 : u<v<w} \frac{h(u,v,w)}{g(u,v,w)} dudvdw \\
&= \iiint_{(u,v,w)\in(c,+\infty)^3 : u<v<w} \lim_{k\to+\infty, l\to 0} \frac{h(u,v,w)}{g(u,v,w)} dudvdw \\
&= \iiint_{(u,v,w)\in(c,+\infty)^3 : u<v<w} \lim_{k\to+\infty, l\to 0} 0^+ dudvdw \\
&= 0^+.
\end{aligned} \tag{18}$$

Hence, for sufficiently large k, we have

$$\lim_{l\to 0} \frac{d\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_1/\mathcal{D}}}{d\,l} = \lim_{l\to 0} -\frac{dS_1}{d\,l} < 0. \tag{19}$$

Thus, the proof is completed. □

B Additional Experiment Results

B.1 Additional Experiment of Language Models' *Resistance*

To broadly validate the phenomenon of *Resistance* in language models, we extend the experimental setup described in Section 5.1. Specifically, we use Llama2-7B (Touvron et al., 2023) as the base model and perform a finer-grained snapshot division within ***forward alignment***, covering a wider range of k and l. As the evaluation metric, we measure the change in relative KL divergence between the distributions of model $\boldsymbol{\theta}_l$ and $\boldsymbol{\theta}_k$, obtained by applying ***forward alignment*** and ***inverse alignment*** to slices $\boldsymbol{\theta}_k$ and $\boldsymbol{\theta}_l$, respectively, relative to the original distributions of $\boldsymbol{\theta}_l$ and $\boldsymbol{\theta}_k$. The experimental results, which is shown in the Table 2, demonstrate that the KL divergence under ***inverse alignment*** is substantially smaller than that under ***forward alignment***, consistent with the conclusions presented in the main text.

Training Steps	$\boldsymbol{\theta}_1$ *vs.* $\boldsymbol{\theta}_2$	$\boldsymbol{\theta}_1$ *vs.* $\boldsymbol{\theta}_3$	$\boldsymbol{\theta}_1$ *vs.* $\boldsymbol{\theta}_4$	$\boldsymbol{\theta}_1$ *vs.* $\boldsymbol{\theta}_5$	$\boldsymbol{\theta}_1$ *vs.* $\boldsymbol{\theta}_6$	$\boldsymbol{\theta}_2$ *vs.* $\boldsymbol{\theta}_3$	$\boldsymbol{\theta}_2$ *vs.* $\boldsymbol{\theta}_4$
Forward Alignment	0.4568 ↑	0.4932 ↑	0.5929 ↑	0.4439 ↑	0.4109 ↑	0.3778 ↑	0.5580 ↑
Inverse Alignment	0.2796 ↓	0.2863 ↓	0.3960 ↓	0.3995 ↓	0.3412 ↓	0.2752 ↓	0.3742 ↓

Table 2: **KL Divergence between different snapshots.** The results show that ***forward alignment*** exhibits higher values compared to ***inverse alignment***, providing evidence for the existence of *Resistance.*

B.2 Additional Experiment of Language Models' *Rebound*

Ablations on the Different Alignment Algorithms. To verify that the consistent *rebound* phenomenon occurs in different alignment algorithms in language models, we conduct language models' *rebound* experiments on reinforcement learning from human feedback (RLHF) (Ouyang et al., 2022), direct preference optimization (DPO) (Rafailov et al., 2024), Kahneman-Tversky optimization (KTO) (Ethayarajh et al., 2024), and simple preference optimization (SimPO) (Meng et al., 2024), using an experimental setup similar to the SFT validation described in Section 5.2. Considering that reinforcement learning algorithms cannot be directly applied to pretrained models, our experimental procedure consists of the following two

steps: (a) SFT of the pretrained model using positive sample data of varying scales as ***forward lignment***; (b) ***inverse alignment*** by applying the corresponding reinforcement learning algorithms on the aligned models, using negative sample data of varying scales, where negative samples serve as selected responses and positive samples serve as rejected responses.

We conduct experiments using the Llama2-7B model on the IMDb task (Maas et al., 2011) with four different RL alignment algorithms. The results, shown in Figure 7, indicate that regardless of the alignment algorithm employed, the model exhibits a decline in performance consistent with that observed in SFT as the amount of positive data increases. Moreover, the rate of this performance degradation accelerates with the increasing volume of positive data. The results further confirm that the broad applicability of language models *rebound* across different alignment algorithms.

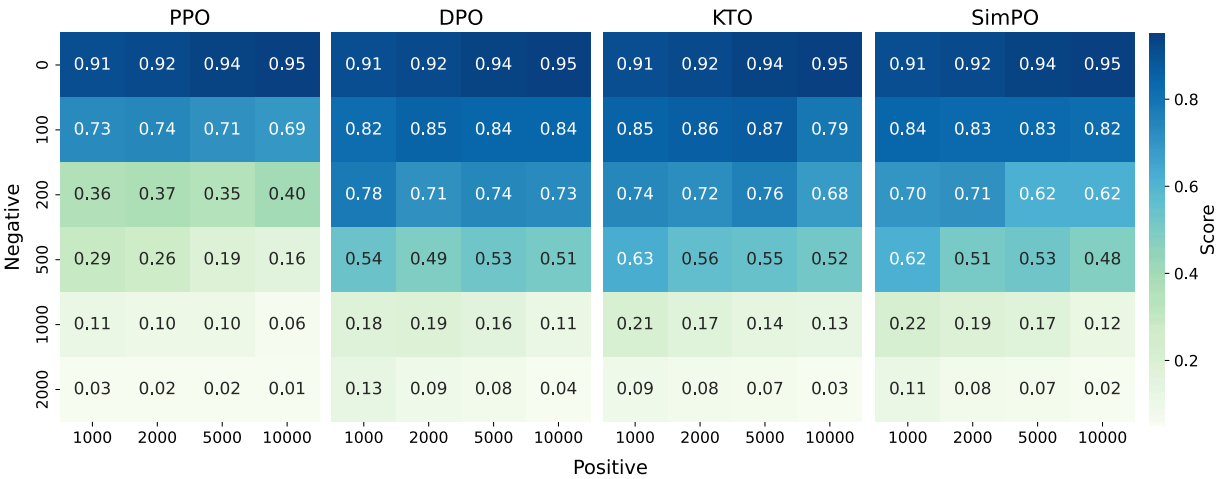

Figure 7: **Experimental results for validating *rebound* across different alignment algorithms.** The heatmaps from left to right represent the performance of the PPO, DPO, KTO, and SimPO algorithms after fine-tuning with varying amounts of positive and negative data. Darker colors indicate better model performance in the heatmaps,. Models trained with more **positive** data perform better initially tend to exhibit worse performance after fine-tuneing with **negative** data. The conclusion holds regardless of the alignment algorithm used.

Ablations on the Different Evaluation Metrics. In the main experiments discussed in Section 5.2, we primarily adopt specific evaluation scores to evaluate the *rebound* phenomenon. Using the score models corresponding to alignment targets as the evaluation metrics is the most direct and important method for the task evaluation. However, to further reinforce the robustness of our findings and address the generality of the *rebound* phenomenon, we acknowledge the value of incorporating broader distribution-level metrics beyond task-specific scoring models. In particular, we consider the Kullback-Leibler divergence between aligned models and base models as an additional evaluation metric.

Specifically, we conducted the evaluation through the following procedure: a) We first measure the KL divergence between the aligned model and the base model before and after fine-tuning the base model with safety data. b) We then apply fine-tuning with unsafe data until the KL divergence between the models decreased to a sufficiently small value ϵ (where $\epsilon = 1 \times 10^{-2}$), and record the amount of unsafe data required as an indicator of the difficulty for the model reverting to its pre-trained distribution. The experimental results are reported in Table 3. Under different models and data scales, we observe that only a small amount of unsafe data is needed for a positively fine-tuned model to revert to the pre-training distribution in terms of KL divergence. Furthermore, the larger the amount of safe fine-tuning, the less unsafe data are required. This observation is consistent with the conclusions drawn in Figure 4.

Ablations on the Reverse Finetuning Settings. To rule out the influence of positive data on the *rebound* phenomenon in language models, we conducted a reverse experimental setup: negative data are used during the SFT stage, while positive data are applied during the ***inverse alignment*** stage. The experimental results, presented in Figure 8, demonstrate that *elasticity* in language models persists under this reverse setting, exhibiting a consistent trend where larger model sizes correspond to greater *elasticity*. The